

Q.8b (15 marks): Erect posture and bipedal locomotion in human evolution

8. (b)

“नृसंस्थाने और कुशल द्विपाद गमन को उपलब्ध मानव विकास का एक अत्यंत महत्वपूर्ण चरण है।” विवचना कीजिए।

"The achievement of erect posture and efficient bipedal locomotion is a critically important stage of human evolution." Discuss.

15

इस हार्निंग में नहीं लिखना चाहिए
Candidate must not write on this margin

The first evidence of erect posture and starting of bipedal locomotion was in the form of Australopithecines that occurred 2-4 mya in Plio-Pleistocene epoch.

THEORIES ABOUT ORIGIN OF ERECT POSTURE

- 1) Optimal Foraging Theory → erect posture leading to free hands was beneficial for optimal foraging.
- 2) Emergence of Grassland habitat → As the ice melted, grasslands emerged which favored the development of bipedal locomotion.

CHANGES THAT ENABLED BIPEDAL LOCOMOTION

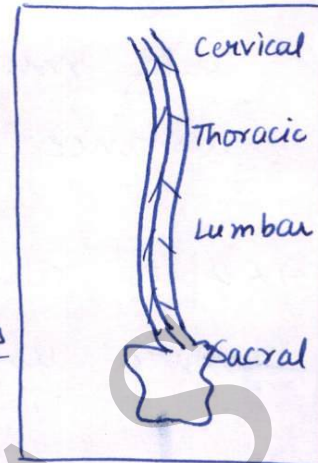
- 1) Foramen Magnum got more forwardly

placed.

2) Vertebral column evolved to have 4 curves

3) Pelvis got shortened and broadened in the form of basin shape.

4) Knees straightened and arches in foot developed.



SIGNIFICANCE OF BIPEDAL LOCOMOTION :-

1) Greater efficiency in walking — 60-70% more efficiency as compared to quadrupedalism.

2) Freed hands — led to development of tools and artefacts.

3) Development of Culture and Society

Bipedal locomotion → Shortened pelvis → smaller gestation period → Greater infant dependency

Institutions like marriage, family ← Formation of mother child bond

4) Safety from Predator → bipedal locomotion led to greater survival as predator could now be visible from a longer distance.

⇒ Despite all these advantages, erect posture also came with FEW CHALLENGES :-

- 1) Frequent Pain in the back, knees due to greater pressure on lower body
- 2) Painful birthing process due to pelvis shortening.
- 3) Efficiency in Running suffered a loss.

Yet; the advantages of bipedalism outnumber the limitations; otherwise human wouldn't have evolved to bipedalism.

Hence, bipedal locomotion and erect posture marked a critical stage in human evolution as it marked human's separation from other primates